DIPLOMA PROGRAMME PRACTICE / SL / SET B

Economics / Paper 2

Worked mark scheme · 105 minutes · 40 marks

Choose one of the two cases and answer all parts (a) to (g) of that case. A calculator is permitted. Use the extracts and data in your answers. Show workings and units, and draw fully labelled diagrams on separate answer paper.

Judge extended responses as a whole using the markband for that part. Award valid alternative analysis and diagrams. In linked calculations, credit correct use of an earlier result. The suggested judgements are examples.

HOW TO MARK THIS PAPER

Use the question-specific marking notes for parts (a) to (f). Apply the 15-mark scale to part (g). Only the selected case contributes to the score.

15-MARK SCALE

0: no relevant economics. 1–3: assertions or isolated concepts. 4–6: some relevant explanation, limited application. 7–9: coherent analysis with some application and consideration of consequences. 10–12: developed, applied analysis weighing competing considerations with a supported view. 13–15: precise theory and relevant diagrams where useful, evidence integrated throughout, and a balanced judgement that explains conditions or relative importance. Other defensible conclusions can receive full credit.

QUESTIONS IN THIS SCHEME

1 Solar power and grid access in India [40]

2 Tourism receipts and inclusion in Jamaica [40]

QUESTION 1 / 40 MARKS

Solar power and grid access in India

Background reading. IEA, India Renewable Energy Market Report 2026

PART a(i) / 2 MARKS

Define a subsidy in the context of the production proposal.

Award. Financial support from government to a producer (1) that reduces its effective cost or increases the return per unit supplied (1). A transfer to the grid operator for a different purpose is not the per-unit payment described.

PART a(ii) / 2 MARKS

Define investment as a component of aggregate demand.

Award. Spending on capital goods (1) that add to or replace productive capacity (1). Buying existing shares does not directly create this investment.

PART b(i) / 3 MARKS

Calculate electricity delivered before and after the grid upgrade, and the percentage increase relative to initial delivery.

Working. Before: 500,000 − 100,000 = 400,000 MWh (1). After: 500,000 − 30,000 = 470,000 MWh (1). Increase = 70,000/400,000 × 100 = 17.5% (1).

PART b(ii) / 2 MARKS

Calculate the annual value of generation displaced by the additional delivery under Text C's assumptions.

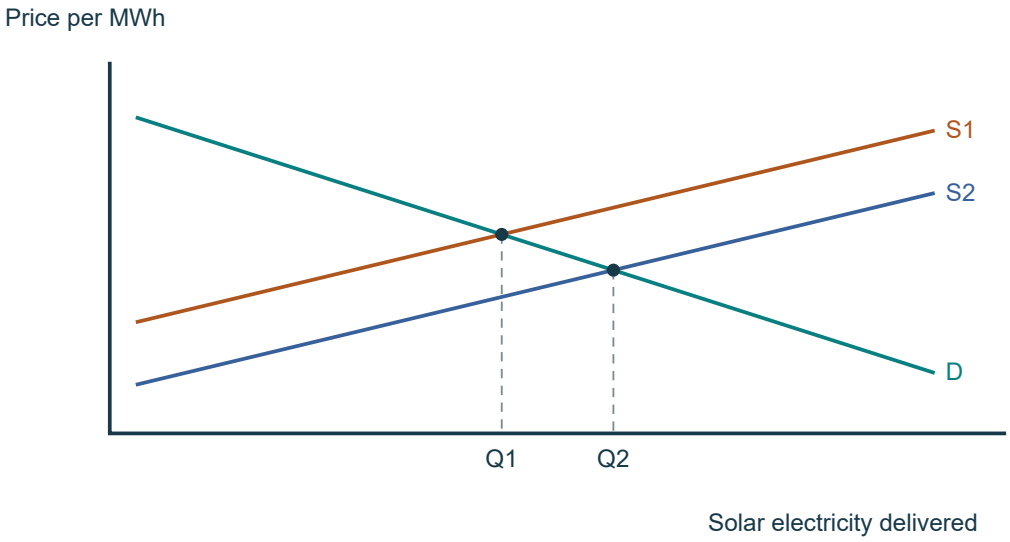
Working. 70,000 MWh × 1,500 rupees/MWh (1) = 105 million rupees per year (1). Credit a consistent own figure from the previous part.

PART c / 4 MARKS

Using a market diagram, explain how the production subsidy should affect the quantity of solar power delivered where the grid has spare capacity.

Diagram and explanation. On delivered solar price and quantity axes shift supply down/right by the subsidy. Mark original and new quantities and the gap between the price paid and price received (up to 2). A higher return per unit encourages production; with a responsive buyer and spare grid capacity equilibrium delivery rises (up to 2).

WORKED REPRESENTATION / c



PART d / 4 MARKS

Explain why a subsidy for additional installed capacity could yield smaller environmental benefits than anticipated in Text A.

Explanation. Installed capacity is potential generation. Midday curtailment prevents part of it from replacing fossil generation. Some accepted generation might displace another clean source. Environmental gain depends on actual marginal generation displaced; evening demand persists. Reward linked analysis grounded in the constraint.

PART e / 4 MARKS

Using an AD–AS diagram, explain a possible long-run effect of more reliable grid infrastructure on industrial output.

Diagram and explanation. Label price level and real GDP and shift the long-run supply curve right, or extend the full-employment part of a Keynesian curve (up to 2). Fewer outages reduce idle machine time and production losses, so firms can sustain higher output with existing inputs. A temporary construction job is not the long-run mechanism (up to 2).

PART f / 4 MARKS

Explain why 450 construction jobs cannot by themselves establish an increase of 450 in total employment.

Explanation. Gross jobs count workers hired for the project. Some are already employed elsewhere; competing projects can lose workers or funding. A net increase depends on unemployment and spare skills. Construction posts also expire, so a one-off count does not measure permanent employment.

PART g / 15 MARKS

Using the texts, data and your knowledge of economics, evaluate whether the state should prioritise grid investment over a production subsidy.

Worked argument. The grid upgrade adds 70,000 MWh of delivered power at unchanged gross generation. The assumed saving is 105 million rupees a year. Dividing the 600 million cost by that saving gives a simple payback of about 5.71 years before discounting, maintenance or environmental gains. More reliable service can also support industrial users.

Counterargument. A delivered-power subsidy can reward actual delivery and attract investment at less congested sites. It is weaker at a binding bottleneck where more panels cannot reach buyers. A uniform payment may favour firms with cheap credit. A single public upgrade faces construction delays and may serve few consumers if the forecast is wrong.

Supported judgement. Prioritise the upgrade if curtailment persists, the asset lasts sufficiently long and displaced generation is genuinely costly. Target any production incentive at times or places of grid value. Evidence about evening storage, maintenance, alternative network routes and who pays could reverse the investment ranking.

Using Figure 1. The illustrative curtailment rates are 5% in the morning, 32% around midday and 8% later in the day. These percentages have different generation denominators: an unweighted mean is not the annual system rate.

Apply the 15-mark scale on the first page to this response as a whole.

QUESTION 2 / 40 MARKS

Tourism receipts and inclusion in Jamaica

Background reading. IMF, Jamaica Article IV consultation (2025)

PART a(i) / 2 MARKS

Define economic development.

Award. A sustained improvement in people's living standards or well-being (1), involving qualitative dimensions such as health, education, opportunity or distribution (1). An increase in GDP alone is insufficient.

PART a(ii) / 2 MARKS

Define a leakage from the circular flow of income and give one example from Text A.

Award. Income withdrawn from the domestic expenditure stream (1). Spending on imported food is one example (1); accept a valid explanation of imports from the extract.

PART b(i) / 3 MARKS

Calculate gross visitor receipts in each year and their percentage growth.

Working. 400,000 × US$900 = US$360 million (1); 480,000 × US$850 = US$408 million (1). Growth = 48/360 × 100 = 13.33% (1).

PART b(ii) / 2 MARKS

Calculate the change in visitor receipts after subtracting the imported inputs in the table.

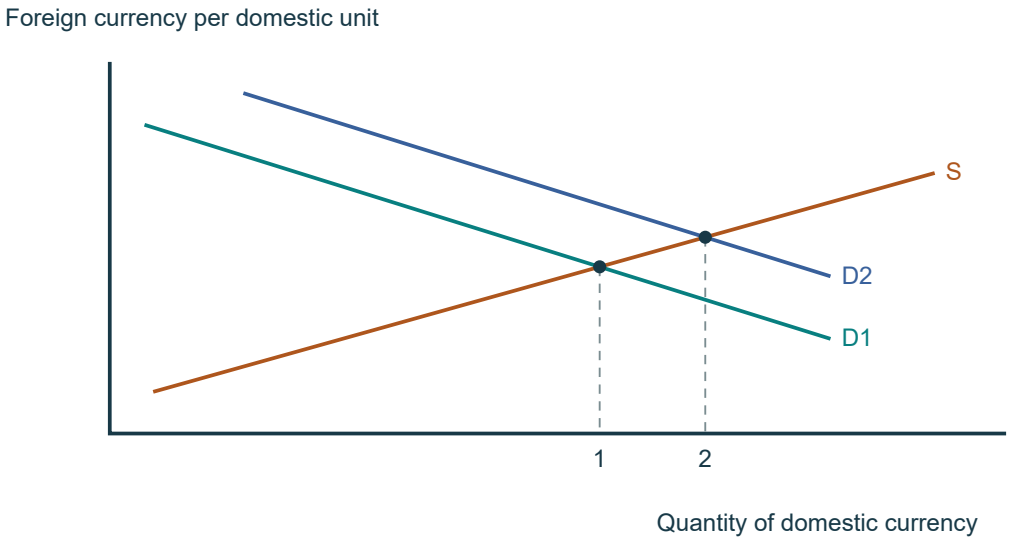
Working. Year 1: 360 − 108 = US$252 million; Year 2: 408 − 144 = US$264 million (1 for consistent working). Increase = US$12 million (1). This is not GDP or the complete current-account balance.

PART c / 4 MARKS

Using a foreign exchange diagram, explain the likely effect of additional foreign visitor spending on the Jamaican dollar in a floating exchange market, other things equal.

Diagram and explanation. Quote foreign currency per Jamaican dollar and use quantity of Jamaican dollars on the horizontal axis. Shift demand for Jamaican dollars right; show appreciation (up to 2). Visitors or their payment intermediaries exchange foreign currency for local goods and services. An equivalent correctly inverted quotation earns credit (up to 2).

WORKED REPRESENTATION / c



PART d / 4 MARKS

Explain why increased demand for hotel meals might lead to only a small increase in local farm incomes in the short run.

Explanation. Long-term import contracts limit short-run substitution. Cold storage and certification constrain local supply even when hotels want fresh produce. Revenue can initially flow to overseas suppliers rather than local farms. Credit linked reasoning about timing and the conditions for a later local gain.

PART e / 4 MARKS

Using a negative externality diagram, explain why the unregulated number of beach visits may exceed the socially efficient number.

Diagram and explanation. Define a visit as the unit. Draw demand and private marginal cost and put marginal social cost above private cost for crowding or damage imposed on others. Mark Qsocial below Qmarket (up to 2). The private decision leaves out costs to fishers or residents; extra visits between these quantities cost society more than they benefit users (up to 2). Accept a consistent external cost of consumption diagram.

PART f / 4 MARKS

Explain why the change in permanent hotel employment is insufficient evidence of a comparable improvement in household welfare.

Explanation. Permanent jobs rose by 150, much less proportionately than arrivals. Job count omits real pay, distribution, insecure seasonal work and environmental costs. More jobs can raise welfare, but household income and access to shared resources matter too. Credit a linked comparison using the context.

PART g / 15 MARKS

Using the texts, data and your knowledge of economics, evaluate whether co-financing local food supply chains would support more inclusive development than additional tourism marketing.

Worked argument. Arrivals rose by 20%, but lower spending per visitor restricted receipts growth to 13.33%. After imported inputs the increase was only US$12 million. Local cold storage can reduce waste and allow farms to compete for hotel contracts, spreading income to rural workers. It can serve residents even when visitor demand falls.

Counterargument. Infrastructure may be underused unless quality, delivery schedules and contract terms suit hotels. Title requirements could concentrate gains among established owners. Marketing can quickly support employment when there is slack, but an extra visitor can increase imported input demand and pressure a crowded beach. The partial accounts cannot establish total local value added.

Supported judgement. Where arrivals have recovered and procurement is the bottleneck, favour shared storage with access for small or tenant growers and measurable procurement targets. Where rooms are empty through weak demand, limited marketing may yield quicker gains. Test progress using farm incomes, local purchases, wages and environmental costs, not arrivals alone.

Using Figure 1. Illustrative local procurement shares are 66% for small hotels, 46% for medium hotels and 27% for large hotels. The gap identifies a possible market for farms; neither causation nor the volume of possible new orders follows from these shares alone.

Apply the 15-mark scale on the first page to this response as a whole.